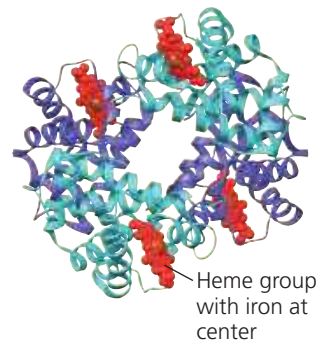


of vertebrates is hemoglobin (see Concept 42.4). In vertebrates, it is contained in erythrocytes and has four subunits. Each consists of a polypeptide and a heme group, a cofactor that has an iron atom at its center (**Figure 42.30**). Each iron atom binds one molecule of O_2 , so a hemoglobin molecule can carry four O_2 molecules. Like all respiratory pigments, hemoglobin binds O_2 reversibly, loading O_2 in the lungs or gills and unloading it elsewhere in the body. This process is enhanced by cooperativity between the hemoglobin subunits (see Concept 8.5). When O_2 binds to one subunit, the others change shape slightly, increasing affinity for O_2 . When four O_2 molecules are bound and one subunit unloads its O_2 , the other three subunits more readily unload O_2 , as an associated change in shape lowers their affinity for O_2 .

▼ **Figure 42.30** Ribbon model of hemoglobin.



Cooperativity in O_2 binding and release is evident in the dissociation curve for hemoglobin (**Figure 42.31a**). Over the range of P_{O_2} where the dissociation curve has a steep slope, even a slight change in P_{O_2} causes hemoglobin to load or unload a substantial amount of O_2 . The steep part of the curve corresponds to the range of P_{O_2} found in body tissues. When cells in a particular location begin working harder—during exercise, for instance— P_{O_2} dips in their vicinity as the O_2 is consumed in cellular respiration. Because of subunit cooperativity, a slight drop in P_{O_2} causes a relatively large increase in the amount of O_2 the blood unloads.

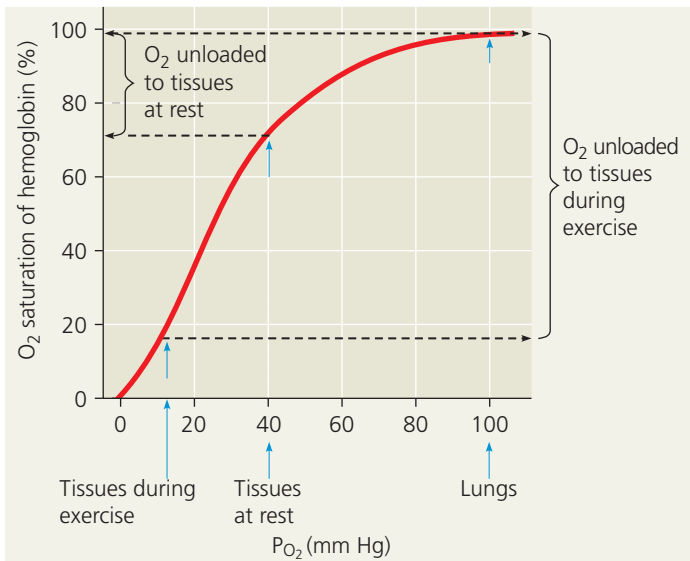
Hemoglobin is especially efficient at delivering O_2 to tissues actively consuming O_2 . However, this increased efficiency results not from O_2 consumption, but rather from CO_2 production. As tissues consume O_2 in cell respiration, they also produce CO_2 . As we have seen, CO_2 reacts with water, forming carbonic acid, which lowers the pH of its surroundings. Low pH decreases the affinity of hemoglobin for O_2 , an effect called the **Bohr shift** (**Figure 42.31b**). Thus, where CO_2 production is greater, hemoglobin releases more O_2 , which can then be used to support more cellular respiration.

Hemoglobin also assists in buffering the blood—that is, preventing harmful changes in pH. In addition, it has a minor role in CO_2 transport, the topic we'll explore next.

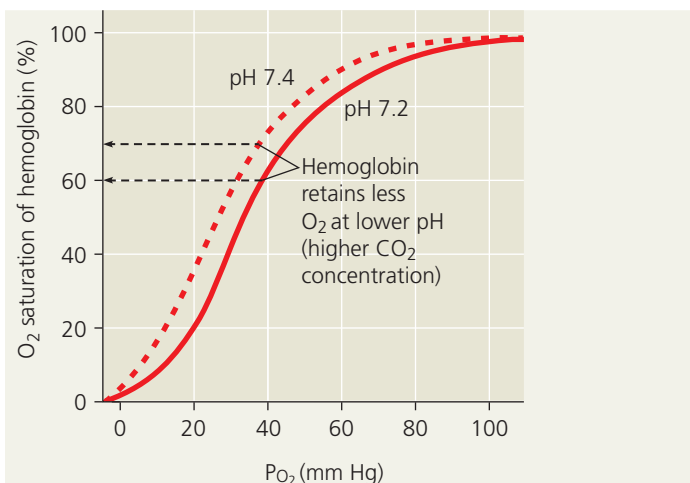
Carbon Dioxide Transport

Only about 7% of the CO_2 released by respiring cells is transported in solution in blood plasma. The rest diffuses from plasma into erythrocytes and reacts with water (assisted by the enzyme carbonic anhydrase), forming H_2CO_3 . The H_2CO_3 readily dissociates into H^+ and HCO_3^- . Most H^+ binds to hemoglobin and other proteins, minimizing change in blood pH. Most HCO_3^- diffuses out of the erythrocytes and is

▼ **Figure 42.31** Dissociation curves for hemoglobin at 37°C.



(a) **P_{O_2} and hemoglobin dissociation at pH 7.4.** The curve shows the relative amounts of O_2 bound to hemoglobin exposed to solutions with different P_{O_2} . At a P_{O_2} of 100 mm Hg, typical in the lungs, hemoglobin is about 98% saturated with O_2 . At a P_{O_2} of 40 mm Hg, common in resting tissues, hemoglobin is about 70% saturated, having unloaded nearly a third of its O_2 . As shown in the above graph, hemoglobin can release much more O_2 to metabolically very active tissues, such as muscle tissue during exercise.



(b) **pH and hemoglobin dissociation.** In very active tissues, CO_2 from cellular respiration reacts with water to form carbonic acid, decreasing pH. Because hydrogen ions affect hemoglobin shape, a drop in pH shifts the O_2 dissociation curve toward the right (the Bohr shift). For a given P_{O_2} , hemoglobin releases more O_2 at a lower pH, supporting increased cellular respiration.

➡ Mastering Biology Animation: Transport of Respiratory Gases

transported to the lungs in the plasma. The remaining HCO_3^- , representing about 5% of the CO_2 , binds to hemoglobin and is transported in erythrocytes.

When blood flows through the lungs, the relative partial pressures of CO_2 favor the net diffusion of CO_2 out of the blood. As CO_2 diffuses into alveoli, the amount of CO_2 in the